



RAN-2411000903011001

S.Y. BCA (DBHUP) (Sem.-III) Examination November - 2025 Database Handling Using Python-Major

Time: 1 Hour]

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book
- (2) All Questions are compulsory.

Seat No.:

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Student's Signature

Q.1.

8

(A) Answer the following Multiple Choice Questions (MCQ) (Any Four):

4

1. Which of the following dot command is used to display the table structure?
 - a. .show
 - b. .header
 - c. .mode
 - d. .schema
2. Which of the following command is used to import CSVdata into SQLite?
 - a. .csv
 - b. .import
 - c. .sqlite
 - d. .tables
3. Which of the following method is used to perform SQLite queries through Python?
 - a. sqlite3()
 - b. execute()
 - c. connect()
 - d. query()
4. Which of the following file access mode is used to write the data at the end of the file?
 - a. read
 - b. write
 - c. append
 - d. All of above
5. Which of the following module is used to visualize data in graphical form?
 - a. matplotlib
 - b. numpy
 - c. pandas
 - d. sqlite

(B) Answer the following questions in short (Any Four):

4

1. List out SQLite advantages.
2. What is the use of .output command in SQLite?
3. What is Package in Python?
4. Which dataframe function is used to display few rows from the top of the dataframe?
5. What is the use of subplot() method?

Q.2. Do as Directed:

9

- (A) Write a note on Trigger.

5

OR

- (A) Explain “Conditional Logic” with example with respect to SQLite.
- (B) “Is it possible to take backup of SQLite database?” Justify your answer.

5

4

Q.3. Answer the following Questions (Any Two):

8

1. Explain connect() and fetchall() methods provided by sqlite module.
 2. What is Dataframe? Discuss “How can you create Dataframe from CSV file?”
 3. Explain plot() method provided by MATPLOTLIB library.
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RAN-2411000903011002 - A

B.C.A. (Sem. - III) Examination December - 2025

303 : Database Handling Using Python (PR)

Time: 2 Hours]

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book.
- (2) Figures to the right indicate full marks of the question.
- (3) All questions are compulsory

Seat No.:

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Student's Signature

Q. 1. Write a Python script to perform the following operations on Transport. [10]

1. Transport(VehicleID, VehicleType, DriverName, Route, Charges)
2. Insert at least 8 records.
3. Display all vehicles operating on "Route-A".
4. Update Charges by +500 for all "Bus" VehicleType.
5. Draw a pie chart showing Charges distribution by VehicleType.

Q. 2. Create table Restaurant(OrderID, Customer, Dish, Quantity, BillAmount) and add at least 6 records. [10]

Write a Python code to do the following:

1. Export the table into an CSV file restaurant.xlsx.
2. Retrieve all orders where Quantity > 3.
3. Drop the table Restaurant.
4. Import the CSV data back into a new table Restaurant_New.

Q. 3. Viva-voce [05]



RAN-2411000903011002 - B

B.C.A. (Sem. - III) Examination December - 2025

303 : Database Handling Using Python (Practical) (PR)

Time: 2 Hours]

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book.
- (2) Figures to the right indicate full marks of the question.
- (3) All questions are compulsory

Seat No.:

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Student's Signature

Q. 1. Write a Python script to perform the following operations on Cinema. [10]

1. Cinema(MovieID, MovieName, Genre, TicketsSold, Collection)
2. Insert at least 10 records.
3. Display all movies belonging to "Comedy" Genre.
4. Delete movies where TicketsSold < 50.
5. Draw a bar chart of Collection for each Genre.

Q. 2. Create table University(ProfID, ProfName, Department, Experience, Salary) and add at least 7 records. [10]

Write a Python code to do the following:

1. Export the table into CSV format university.csv.
2. Retrieve professors having more than 10 years of Experience.
3. Drop the table University.
4. Import csv data back into a new table University_New.

Q. 3. Viva-voce [05]



RAN-2411000903011002 - C

B.C.A. (Sem. - III) Examination December - 2025

303 : Database Handling Using Python (PR) (Practical)

Time: 2 Hours]

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book.
- (2) Figures to the right indicate full marks of the question.
- (3) All questions are compulsory

Seat No.:

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Student's Signature

- Q. 1.** Write a Python script to perform the following operation on Movie. [10]
1. Create table Movie(MovieId, Title, Director, Genre, ReleaseYear)
 2. Insert atleast 10 records.
 3. Display all movies where Genre is 'Action' and ReleaseYear is between 2015 and 2021.
 4. Count the number of movies available in each Genre.
 5. Generate a Pie chart to show the number of movies per Genre.
- Q. 2.** Create a table named Library(BookID, Title, Author, Category, Price) and insert atleast 5 records. [10]
- Write the python code to do the following:
1. Export the entire table data into a CSV file named library.csv.
 2. Write a query to display all books where Price is greater than 500.
 3. Display all books whose name starts with 'B'.
 4. Dump the Library table structure into Library.sql file.
- Q. 3.** Viva-voce [05]



RAN-2411000903011002 - D

B.C.A. (Sem. - III) Examination December - 2025
303 : Database Handling Using Python (Practical)

Time: 2 Hours]

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book.
- (2) Figures to the right indicate full marks of the question.
- (3) All questions are compulsory

Seat No.:

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Student's Signature

- Q. 1. Create database with following tables. [10]**
- A. Cust_Master (Cid, CName, ContactNo, City)
Prod_Master (Pid, PName, Company, Price, Qty)
 - B. Create Python Program to display data of Prod_Master Table.
- Q. 2. Write a python script perform following task [10]**
- 1) Add the column email_id in Cust_Master table.
 - 2) Export CSV file of All Customers data.
- Q. 3. Viva [05]**



RAN-2411000903011002 - E

B.C.A. (Sem. - III) Examination December - 2025

303 : Database Handling Using Python (PR) (Practical)

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book.
- (2) Sketch neat and labelled diagram wherever necessary.
- (3) Figures to the right indicate full marks of the question.
- (4) All questions are compulsory

Seat No.:

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Student's Signature

- Q. 1.** Write a python code to implement following operation on Employee table using SQLite module. [10]
Employee (emp_id, emp_name, contact_no, department_name, DOJ)
- A. Insert 10 records.
 - B. Display all records using cursor.
- Q. 2.** Write a menu driven python script perform following task [10]
- 1) Display the details of sales department.
 - 2) Display details of employee working with emp_id is 105.
- Q. 3.** Viva [05]



RAN-2411000903011002 - F

B.C.A. (Sem. - III) Examination December - 2025

303 : Database Handling Using Python (PR) (Practical)

Time: 2 Hours]

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book.
- (2) Figures to the right indicate full marks of the question.
- (3) All questions are compulsory

Seat No.:

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Student's Signature

- Q. 1.** Consider a "product.csv" file columns pid, pname, category, price. [10]
1. Using writer object, write 5 records in product.csv.
 2. Read data using reader object.
 3. Insert all record into SQLite table "Product" using cursor object.
 4. Using cursor, display all record where price > 70.
- Q. 2.** Create a text file data.txt and perform the following tasks: [10]
1. Write few lines into the file.
 2. Read the file.
 3. Count the total number of lines.
 4. Display the first line of the file..
- Q. 3.** Viva-voce [05]



RAN-2411000903011002 - G

B.C.A. (Sem. - III) Examination December - 2025

303 - Database Handling Using Python (PR) (Practical)

Time: 2 Hours]

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book.
- (2) Figures to the right indicate full marks of the question.
- (3) All questions are compulsory.

Seat No.:

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Student's Signature

Q. 1. Write a Python script to perform the following operations on Orders. (10)

1. Orders(OrderID, CustomerName, Product, Quantity, Amount)
2. Insert at least 10 records.
3. Display all orders whose Amount is greater than 5000.
4. Delete all orders where Quantity < 2.
5. Draw a line chart showing Amount spent by each Customer.

Q. 2. Create table School(StudentID, Name, Class, Marks, Age) and add at least 7 records. Write a Python code to do the following:

1. Export the table data into a CSV file named school.csv.
2. Retrieve all students whose Marks are greater than 80.
3. Delete the table School.
4. Import the data back from CSV into a new table School_New.

Q. 3. Viva-voce (05)



RAN-2411000903011002 - H

B.C.A. (Sem. - III) Examination December - 2025

303 - Database Handling Using Python (PR) (Practical)

Time: 2 Hours]

[Total Marks: 25

સૂચના : / Instructions

- (1) ઉપરોક્ત દર્શાવેલ વિગતો ઉત્તરવહી પર અવશ્ય લખવી.
Fill up strictly the above details on your answer book.
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- (3) All questions are compulsory

Seat No.:

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Student's Signature

Q. 1. Write a Python script to perform the following operations on Sales. (10)

1. Sales(SalesID, Item, Region, Quantity, Revenue)
2. Insert at least 10 records.
3. Display all sales made in the "North" Region.
4. Update Revenue = Revenue + 1000 for all sales where Quantity > 20.
5. Draw a bar chart of total Revenue by Region.

Q. 2. Create table Clinic(DoctorID, DoctorName, Specialization, Patients, Fees) and add at least 6 records.

Write a Python code to do the following:

1. Export the table into an CSV file clinic.csv.
2. Retrieve doctors whose Specialization is "Cardiology".
3. Drop the table Clinic.
4. Import the Excel data back into a new table Clinic_New.

(10)

Q. 3. Viva-voce

(05)